REVIEW ARTICLE

Sports Med 2011; 41 (2): 103-123
0112-1642/11/0002-0103/$49.95/0

© 2011 Adis Data Information BV. All rights reserved.

Two Emerging Concepts for Elite Athletes

The Short-Term Effects of Testosterone and Cortisol on the Neuromuscular System and the Dose-Response Training Role of these Endogenous Hormones

Blair T. Crewther,[1,2,3] *Christian Cook*,[3,4,5] *Marco Cardinale*,[6,7] *Robert P. Weatherby*[2] and *Tim Lowe*[8]

1 The New Zealand Institute for Plant & Food Research Limited, Hamilton, New Zealand
2 Department of Exercise Science and Sport Management, Southern Cross University, Lismore, New South Wales, Australia
3 Hamlyn Centre, Institute of Global Health Innovation, Imperial College, London, UK
4 United Kingdom Sport Council, London, UK
5 Sport, Health and Exercise Science, Bath University, Bath, UK
6 British Olympic Medical Institute, London, UK
7 University College London, Division of Surgical and Interventional Science, London, UK
8 School of Applied Sciences, Bay of Plenty Polytechnic, Tauranga, New Zealand

Contents

Abstract . . . 104
1. Introduction . . . 104
2. Literature Search Methods . . . 105
3. Hormonal Effects on the Neuromuscular System . . . 105
   3.1 Long-Term Effects of Exogenous Hormones . . . 106
      3.1.1 Muscle Development . . . 106
      3.1.2 Neural Development . . . 106
   3.2 Short-Term Effects of Exogenous Hormones . . . 106
      3.2.1 Steroid Receptors . . . 106
      3.2.2 Second Messengers and Lipid/Protein Pathways . . . 107
      3.2.3 Behaviour and Cognition . . . 107
      3.2.4 Motor System . . . 107
      3.2.5 Muscle Properties . . . 107
      3.2.6 Energy Metabolism . . . 108
   3.3 Summary . . . 108
4. Hormonal Contribution to Resistance Training . . . 109
   4.1 Acute Modifications in Endogenous Hormones . . . 109
      4.1.1 Workout Design . . . 109
      4.1.2 Nutrition . . . 110
      4.1.3 Training Status and Type . . . 110
      4.1.4 Genetic Variation . . . 111
   4.2 Chronic Modifications in Endogenous Hormones . . . 111
      4.2.1 Training Status . . . 111
      4.2.2 Training Type . . . 112
   4.3 Acute and Chronic Modifications in Steroid Receptors . . . 113
      4.3.1 Androgen Receptors . . . 113

4.3.2 Glucocorticoid Receptors 113
4.4 Summary 114
5. Limitations and Challenges with Hormonal Research 114
6. Conclusion 115

Abstract

The aim of this review is to highlight two emerging concepts for the elite athlete using the resistance-training model: (i) the short-term effects of testosterone (T) and cortisol (C) on the neuromuscular system; and (ii) the dose-response training role of these endogenous hormones. Exogenous evidence confirms that T and C can regulate long-term changes in muscle growth and performance, especially with resistance training. This evidence also confirms that changes in T or C concentrations can moderate or support neuromuscular performance through various short-term mechanisms (e.g. second messengers, lipid/protein pathways, neuronal activity, behaviour, cognition, motor-system function, muscle properties and energy metabolism). The possibility of dual T and C effects on the neuromuscular system offers a new paradigm for understanding resistance-training performance and adaptations.

Endogenous evidence supports the short-term T and C effects on human performance. Several factors (e.g. workout design, nutrition, genetics, training status and type) can acutely modify T and/or C concentrations and thereby potentially influence resistance-training performance and the adaptive outcomes. This novel short-term pathway appears to be more prominent in athletes (vs non-athletes), possibly due to the training of the neuromuscular and endocrine systems. However, the exact contribution of these endogenous hormones to the training process is still unclear. Research also confirms a dose-response training role for basal changes in endogenous T and C, again, especially for elite athletes. Although full proof within the physiological range is lacking, this athlete model reconciles a proposed permissive role for endogenous hormones in untrained individuals. It is also clear that the steroid receptors (cell bound) mediate target tissue effects by adapting to exercise and training, but the response patterns of the membrane-bound receptors remain highly speculative.

This information provides a new perspective for examining, interpreting and utilizing T and C within the elite sporting environment. For example, individual hormonal data may be used to better prescribe resistance exercise and training programmes or to assess the trainability of elite athletes. Possible strategies for acutely modifying the hormonal milieu and, thereafter, the performance/training outcomes were also identified (see above). The limitations and challenges associated with the analysis and interpretation of hormonal research in sport (e.g. procedural issues, analytical methods, research design) were another discussion point. Finally, this review highlights the need for more experimental research on humans, in particular athletes, to specifically address the concept of dual steroid effects on the neuromuscular system.

1. Introduction

Resistance training provides an effective stimulus for improving the morphological (e.g. muscle size) and/or functional (e.g. strength, power) qualities of the neuromuscular system in athletic and non-athletic populations.[1-3] The steroid hormones are recognized as one important mediator

© 2011 Adis Data Information BV. All rights reserved.

for these training adaptations. Specifically, the biological effects of testosterone (T) and cortisol (C) help to control long-term changes in muscle growth and subsequent performance, especially with resistance training.[4-6] However, in light of recent advancements in this area, a more complex model may be needed to explain the T and C contribution to resistance-training adaptations.

Greater understanding of the steroid hormones, especially through animal research, has led to suggestions that they can also influence neuromuscular function (e.g. behaviour, neuronal activity, intracellular signalling, muscle force) through various rapid or short-term mechanisms (see reviews[7-10]). Therefore, T and C may well contribute to the adaptive responses to training by regulating long-term muscle performance and/or the short-term expression of neuromuscular performance. To date, little research has addressed the concept of dual steroid effects with the second tier being a possible moderator of human movement during normal everyday and sport-specific activities.

Debate also exists regarding the permissive (i.e. need to be present) and dose-response (i.e. changes need to occur) training roles of these endogenous hormones. Recent studies on untrained males suggest that physiologically elevated hormones do not enhance muscle size and strength.[11-14] However, elite athletes can differ considerably from lesser trained or untrained individuals in terms of both endocrine and neuromuscular function and development,[15-20] thereby accounting for many of the differences and discrepancies found with regards to the training roles of T and C. The possibility of short-term steroid effects offers a further avenue to explain the observed or expected differences between athletes and non-athletes.

This review will highlight two emerging concepts for elite athletes using the resistance-training model: (i) the short-term T and C effects on the neuromuscular system; and (ii) the dose-response training role of these endogenous hormones. Within the athlete model, we will also discriminate between highly trained and trained individuals to emphasize the different hormonal effects and roles. Section 2 outlines the literature search methods. Section 3 uses exogenous evidence to investigate the possibility of dual T and C effects on the neuromuscular system, focusing on potential mechanisms, their time course and target tissue. Section 4 uses endogenous evidence to examine how modifications in T and/or C concentrations may contribute to resistance-training performance and adaptations. This section also examines the exercise and training responses of the steroid receptors. Section 5 highlights some of the limitations and challenges associated with hormonal research.

2. Literature Search Methods

The databases searched were PubMed and Scirus. Literature searches were undertaken using several key words including 'testosterone', 'androgen', 'cortisol', 'glucocorticoid', 'non-genomic', 'rapid', 'androgen receptors', 'glucocorticoid receptors', 'neuronal', 'nervous system', 'behaviour', 'brain', 'neuromuscular', 'muscle', 'resistance training', 'resistance exercise', 'strength training', 'adaptation', 'power', 'strength', 'muscle', 'athlete', 'nutrition', 'supplementation' and 'males'. Following electronic searches, the reference lists of included studies were screened for additional relevant research.

The inclusion criteria for research in section 3 were randomized, placebo-controlled trials (preferably blinded). The inclusion criteria for research in section 4 were less stringent, particularly for those studies involving elite athletes, because of inherent limitations with performing research on athletes and within the elite sporting environment (see section 5). Given the broad nature of this review, other relevant studies and reviews of literature have been included in all sections.

3. Hormonal Effects on the Neuromuscular System

Human movement is controlled by the physiological interactions of the CNS (i.e. brain, spinal cord) and the peripheral nervous system (PNS) [i.e. motor unit-neuron and innervated muscle fibres, collectively termed the neuromuscular system].[21] Briefly, during voluntary movement, neural signals from the brain are transmitted down the spinal cord to activate the

© 2011 Adis Data Information BV. All rights reserved.

motor neurons and stimulate muscle fibre contractions to produce (and coordinate) muscle force for the expression of human movement during normal everyday and athletic tasks.[21,22] Force development also depends upon other morphological, neurological, metabolic and biomechanical factors.[3,21,22] This section will examine the effects of exogenous T and C on some of these factors.

3.1 Long-Term Effects of Exogenous Hormones

3.1.1 Muscle Development

In animals, the most recognized effect of T[23-29] and C[30-33] lies in the remodelling of skeletal muscle protein, especially the type II muscle fibres.[24,25,28,31-33] The influence of T[34-40] and C[41-44] on protein metabolism and muscle size are also well documented in humans, but the fibre-specific effects are less clear,[36,37,44] probably because of differences in experimental design between animal and human studies (e.g. treatment dose and duration, muscles assessed). Nevertheless, T is considered a primary anabolic hormone, by increasing protein synthesis and decreasing protein degradation, and C the primary catabolic hormone, as it increases protein degradation and decreases protein synthesis.[6,45-47] The net result is an increase, decrease or maintenance in muscle fibre and whole muscle size.

The force-generating capacity of muscle is related to its cross-sectional area,[48] so it is generally agreed that T and C contribute to the expression of human performance by regulating longer term (i.e. weeks to months) changes in skeletal muscle size. We do recognize that several other events and pathways (e.g. activation of satellite cells, local growth and mechanical factors, feeding) may contribute to net changes in muscle size, along with other morphological adaptations (e.g. tendon and connective tissue, muscle architecture),[3,45,46] but these will not be addressed in this review.

Resistance training provides an effective stimulus for maximizing muscle form and function. For example, the exogenous manipulation of T in males enhanced muscle size and/or performance in a dose-dependent manner,[34-38,40,49] but combining T with resistance training produced greater adaptation than either alone.[50-52] Likewise, suppressing total T and free T attenuated the training gains in muscle size and strength.[53] Exercise and training can also counteract the catabolic effects of total C and free C.[30,31,33,43,44,54] However, the initial (<8 weeks) force changes with training are primarily ascribed to neural adaptation, after which muscle growth predominates.[55-59] Some of this evidence also supports a permissive hormonal role in regulating overall changes in muscle size and strength, which will be addressed in more detail in section 4.

3.1.2 Neural Development

Skeletal muscle growth is not the only outcome influenced by these steroids. The motor neuron exhibits T-sensitive plasticity in terms of soma size,[29,60] dendritic length[60,61] and synaptic input,[62] along with the size[63] and number[64] of motor neurons. Likewise, different C treatments have been shown to influence dendritic growth,[65,66] soma size[66] and the structural and functional characteristics of the neuromuscular junction.[67] Still, it remains to be seen if these neural modifications can occur independently of the contractile proteins and whether or not exercise and training can enhance, or attenuate, the steroid effects.

3.2 Short-Term Effects of Exogenous Hormones

3.2.1 Steroid Receptors

In general, the long-term effects of T and C are mediated by specific receptors (e.g. androgen receptors [AR] and glucocorticoid receptors [GR]) located in the cytosol of target cells, which regulate the transcription of deoxyribonucleic acid.[8,10,68] The identification of steroid receptors on the membrane of cells provides a new viewpoint for understanding the biological effects of steroid hormones.[10] These newly identified receptors include ion channels, neurotransmitters, protein kinases and G-proteins.[10] The exact role of these receptors has yet to be clearly defined, but they are thought to mediate cellular effects that are independent of the cytosol receptors and

© 2011 Adis Data Information BV. All rights reserved.

gene transcription, with these effects also occurring on a much shorter timescale.[8,10,68]

3.2.2 Second Messengers and Lipid/Protein Pathways

Steroid binding to membrane receptors can facilitate the rapid (i.e. seconds) activation of many second messenger signals, such as the intracellular release of ions (e.g. Ca^{2+}, adenosine triphosphate [ATP], K^{+}) in different cells,[69-76] with downstream effects on various lipid (e.g. inositol trisphosphate, phosphatidylinositol 3-kinase)[72-74,77] and protein kinase (PK) pathways (e.g. mitogen-activated PK, PKC, PKA).[69,70,74,78,79] These modifications in the cellular environment can subsequently affect neuronal excitation or inhibition within minutes, as seen in different brain regions.[80-84] Ultimately, the hormonal induction of these signals, along with changes in the membrane kinetics of cells, could affect the functioning of neuromuscular tissue/systems to which the cells belong. The following sub-sections will address different aspects of neuromuscular function important to human movement and performance.

3.2.3 Behaviour and Cognition

Studies on animals have demonstrated rapid (i.e. minutes) T- or C-induced modifications in behaviour[85-88] that are likely to originate from alterations in brain neuronal activity. Likewise, in humans, long-term users of anabolic steroids often exhibit behavioural traits (e.g. mood disturbances, aggression) that reflect alterations in central neural activity.[89] Research on humans[90-96] and primates[97,98] also confirms a steroidal role in regulating or supporting different aspects of cognitive function (e.g. memory, decision making, learning). The T or C effects on behaviour and cognition do appear to be more delayed in humans, occurring several hours after initial treatment,[90-96] but this is probably due to methodological reasons (e.g. timing of assessment, treatment type) rather than any inherent delay in the transduction processes influencing the CNS.

3.2.4 Motor System

The motor system is another important steroid target. In one study,[99] the motor-evoked potential of muscle increased after the electrical stimulation of the motor cortex when salivary free C concentrations were low, but an oral C dose prevented this response. The free C concentrations of subjects also correlated (negatively) to the motor cortex response. Similarly, an increase in total T concentrations, induced by a single injection of human chorionic gonadotropin, decreased the cortical motor threshold in humans[100] and this can also influence muscle activity. These results are partially supported by human studies where exogenous treatments were found to modify the secretory levels of these hormones and muscle performance.[34,35,39,40,44,49-51,53] Together, the hormonal data presented has possible implications for transmitting and processing neural information to and from skeletal muscle, as well as the putative effects on behaviour and cognition.

The biological effects of these steroids on the CNS are complex. The brain uptake of T involves various distribution routes including cerebrospinal fluid, diffusion through brain tissue and transport through nerve projections.[101] Experimental data also suggests that T secretion is regulated by a direct neural connection between the brain and testes, via the catecholamines.[102-104] Thus, the dual effects of T may be mediated by two pathways, the 'slow-release' hypothalamic-pituitary-gonadal axis and a 'rapid-release' hypothalamic-gonadal pathway. Even then, the target response may depend on the peripheral conversion of T or C to other metabolites (e.g. estradiol, dehydrotestosterone, tetrahydrocortisol)[105-108] or mediated through other neurotransmitters (e.g. serotonin).[109] To complicate matters, many of the androgens and glucocorticoids influencing behaviour and cognition are themselves synthesized within the CNS.[106,107]

3.2.5 Muscle Properties

Exogenous T and C may contribute to the muscular properties of the PNS by rapidly modulating Ca^{2+} influx in skeletal muscle cells.[71-74] These findings are important because intracellular Ca^{2+} not only triggers the muscle contraction, but is also involved in twitch relaxation, energy metabolism and helps to maintain the structural integrity of the muscle fibre.[110] The expression of Ca^{2+} channels in skeletal muscle are also

© 2011 Adis Data Information BV. All rights reserved.

sensitive to T[111] and C[112] treatments. Another mechanism by which steroid hormones may help to regulate muscle contractility is through the acute and chronic modification of NA^+-K^+ pump activity and content, along with other peptides and growth factors.[113]

Animal studies have demonstrated rapid T or C effects on the electrophysiological and contractile properties of muscle (e.g. reflex response, neuromuscular transmission, end plate potential).[114,115] The effects of longer term steroidal treatments on muscle performance have been widely investigated in animals[23,28,116] and humans.[34,35,39,40,44,49-51,53] In humans, the performance gains (or losses) observed were generally proportional to the temporal changes in muscle size, but few studies have examined specific or relative performance to address the prospect of a mass-independent mechanism[44,49] and those that have are with conflicting results. In animals, T manipulation did influence relative forces,[23,28,116] but there was a lack of uniformity across different tests and muscle groups. The lack of experimental research in this area highlights the need for more studies on humans with a specific focus on the dual steroid effects.

3.2.6 Energy Metabolism

The primary role of C as a glucocorticoid is to stimulate gluconeogenesis and glycogenolysis via glycogen, protein and lipid metabolism and through its permissive actions on other hormones (e.g. catecholamines and glucagon).[41,42,47,117,118] Exogenous C can also affect human sympathetic nerve activity within hours,[119,120] which has further consequences for metabolic and cardiovascular function. The metabolic role of T is less clear. T promoted rapid effects on substrate transport and metabolism in cultured myotubes[121] and regulated insulin sensitivity in rats.[122,123] However, no differences in substrate use and endurance performance were found in humans with either low, normal or high total T concentrations.[124] The long-term response to T manipulation is equally unclear with reports of increases, decreases or no changes in various metabolites and substrates involved in the energy-yielding processors.[25-27,125-128]

Long-term T administration has been found to improve the fatigue resistance of specific muscles in rats.[24,116] Hence, the metabolic effects of T may only be realized when muscle fatigue occurs and is more localised. In fact, many of the fatigue mechanisms are sensitive to the rapid actions of this androgen, including neural drive to the motor neurons, neuromuscular propagation, excitation coupling, metabolic substrates and the intracellular milieu.[129] However, an exogenous protocol that modified T levels in humans did not affect fatigability.[49] Once again, differences in research design make comparisons difficult and there is a clear need for more experimental studies on humans.

Research confirms that T and C can influence neuromuscular performance via one or many mechanisms that can be broadly classified as being either a short-term or long-term effect, thereby potentially contributing to human movement (figure 1). It is also clear that these effects are dose dependent, although a permissive role cannot be dismissed. Other issues remain unresolved, such as whether the opposing long-term steroid actions also apply in the short term, or any interactions thereof. Indeed, the rapid steroid actions may prepare target cells for chronic adaptations and induce early adaptive changes that are opposite to the delayed effects.[8,10] Since the use of exogenous hormones has moral, legal and health implications, another key issue is whether the observed responses can occur under physiological conditions (i.e. endogenous hormones). Whether we can utilize the novel short-term steroid pathway to achieve the best performance outcomes, and practical strategies for doing so, are unknown.

3.3 Summary

Exogenous evidence confirms that T and C can regulate long-term changes in muscle growth and performance, especially with resistance training. This evidence also confirms that changes in T or C concentrations can moderate or support the performance capacity of the neuromuscular system through various short-term mechanisms (e.g. second messengers, lipid/protein pathways, neuronal activity, behaviour, cognition, motor system function, muscle properties and energy metabolism).

© 2011 Adis Data Information BV. All rights reserved.

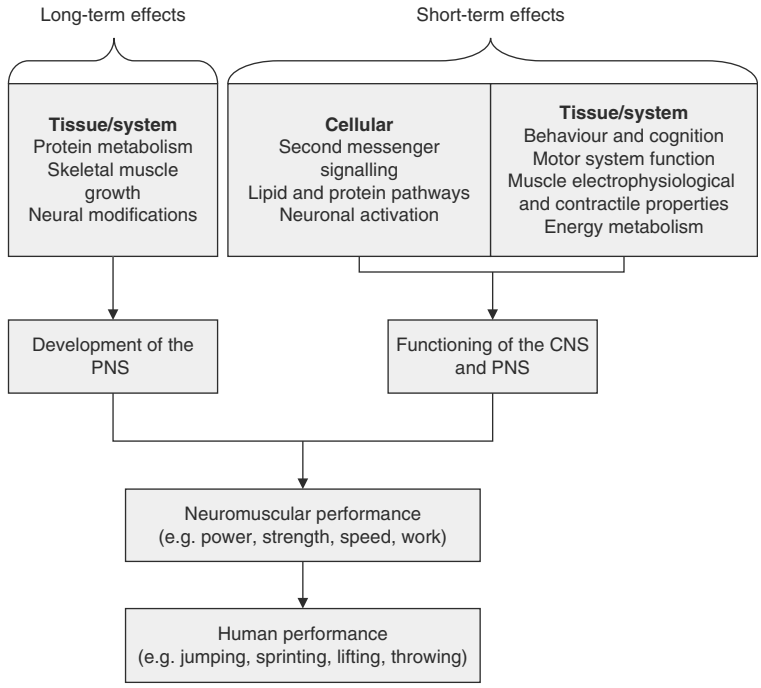


Fig. 1. Summary of the dual effects of testosterone and cortisol on the neuromuscular system and the implications for human performance. **PNS** = peripheral nervous system.

The possibility of dual T and C effects on the neuromuscular system offers a new paradigm for understanding their contribution to resistance-training performance and adaptations.

4. Hormonal Contribution to Resistance Training

4.1 Acute Modifications in Endogenous Hormones

4.1.1 Workout Design

Resistance exercise workouts are prescribed, in part, on the premise that acute elevations in endogenous hormones increase the likelihood of receptor interactions, thereby mediating the long-term adaptive responses.[4,6] For example, hypertrophy workouts produce acute increases in various anabolic (e.g. T, growth hormone, growth factors) and catabolic (e.g. C) hormones to promote protein metabolism and muscle growth during the recovery period.[130-134] Conversely, maximal strength and power workouts typically produce little or no hormonal change,[130-135] so the adaptive changes are generally limited to the neural pathways. This perspective may be too simplistic given the potential for dual steroid effects on the neuromuscular system and human performance outcomes (e.g. power, strength, speed, work), which are major determinants of power and strength adaptation.[136-138]

It is possible that acute changes in physiological hormones may offer short-term benefits for training performance and long-term adaptations. Some studies on athletes have used intense exercise as a stimulus to acutely elevate T concentrations and improve power or strength during later exercises.[139,140] However, this approach did not enhance training adaptation in rugby players,[141] partly due to individual variation in the exercise responses. In untrained males,[142] a protocol that combined arm and leg exercises was also used to elevate the anabolic hormones during individual workouts, thereby resulting in greater training gains in arm strength (vs arm only training).

© 2011 Adis Data Information BV. All rights reserved.

These findings are limited by differences in baseline strength between training groups[142] and the lack of control data.[140]

Recent work on untrained males has found that physiologically elevated hormones do not necessarily enhance protein metabolism, muscle growth or strength.[11-14] Taken in perspective with other exogenous evidence,[34-38,40,49,53] it appears that anabolic hormones may have a permissive effect on improving muscle form and function in untrained males. That is, adaptive changes in muscle size and performance may result from resistance training protocols as long as basal T concentrations are within the physiological range (i.e. T value within this range is less important). However, elite athletes involved in power sports can often express much higher T values,[143-145] so a greater training stimulus may be needed to evoke acute alterations in the hormonal milieu (i.e. dose response) and thereby allow such individuals to achieve further increases in muscle size and performance.

4.1.2 Nutrition

Nutritional adjustments offer another strategy for modifying the hormonal and performance outcomes. One study noted improvements in the power and strength of recreational lifters using a multi-nutrient supplement for 7 days.[146] As a possible mechanism to explain these gains, the total T and free T responses to exercise (along with other anabolic markers) were elevated after supplementation. Further evidence comes from training studies. Carbohydrate and/or protein supplementation has been found effective in acutely elevating T and/or reducing C concentrations during workouts and this appeared to promote greater resistance training gains in body mass, muscle size and performance,[147-149] whilst also attenuating the strength losses that occurred during overreaching.[150]

There are still some conflicting reports regarding the efficacy of carbohydrate and/or protein supplementation. These include acute modifications in the workout responses of T and/or C but without any alterations in training performance,[151-153] no hormonal alterations but improved performance[154,155] or no changes in either of these outcomes.[156-158] These inconsistencies could be explained by differences in the supplementation procedures, workout design and the training background of subjects. Nevertheless, the possibility exists for using different strategies (e.g. workout design, nutrition) to acutely modify the hormonal milieu and thereafter, the performance/training outcomes.

4.1.3 Training Status and Type

Resistance training can enhance the T responses and/or reduce the C responses to workouts in previously untrained males,[57,147,159-161] probably as an early-phase adaptation to assist muscle growth. Strength athletes can also produce greater workout changes in T than non-athletes,[16,17] with more experienced weightlifters producing greater T responses than less experienced lifters.[162] Since strength athletes are already highly adapted to the training stimulus, and given their limited ability (and sometimes desire) to induce further muscle growth, these response patterns could represent a longer term adaptation to support workout performance. This idea is supported by correlations between the T and performance responses to exercise in athletes,[17,139,163-166] but a lack of a correlation in non-athletes.[17]

The type of training performed is an important consideration. Athletes who often train to induce muscle fatigue (e.g. bodybuilders, 400 m sprint runners) have shown little-to-no changes, or a reduction, in T after exercise bouts.[163,167-169] Likewise, the T and C responses of endurance athletes to resistance exercise were less pronounced than resistance-trained athletes.[16] Similar patterns have been noted during other exercise forms. In response to a 400 m sprint,[167] the total T and free T concentrations of elite runners were lowered despite an increase in luteinizing hormone, which stimulates T production. Conversely, non-elite runners had elevated T, but luteinizing hormone levels did not change. These findings have led to speculation of enhanced T uptake and utilization to attenuate muscle fatigue.[6,163]

Literature confirms that acute modifications in these endogenous hormones may have different training roles for highly trained, trained and

© 2011 Adis Data Information BV. All rights reserved.

untrained individuals. The different roles may be likened to the morphological and neural contributions to training adaptation, where one predominates during the early phase and another during later phases.[1-3] It is important to note that strength and power athletes also exhibit ideal neuromuscular profiles for promoting the steroid effects, such as greater type II fibre area and a higher percentage of type II fibres (vs untrained individuals).[19,170,171] Thus, the effects of physiological changes in T or C concentrations may be further linked to the recruitment and training of those neuromuscular structures that are more sensitive to these steroids.[24,25,28,31-33]

4.1.4 Genetic Variation

Genetic variation is a major limiting factor in human movement and may account for up to half of the variation in performance between athletes.[172] In male athletes, large individual variation has also been observed in the T and/or C responses to resistance exercise[141,173,174] and other exercise forms.[140,169,174-176] These individual patterns may reflect genetic differences in our ability to respond to, and cope with, different stressors,[169] and this could be important for moderating physiological adaptation to training stress.[169,173,174,176] Therefore, repeated exposure to the stressors of exercise, training and competition may result in steroid responses akin to our genetic makeup. It is important to note other genetic factors (e.g. body and muscle composition, anatomy, physiology, behaviour) as potential moderators of athlete performance and training.[172]

A recent study of rugby players addressed the prospect of prescribing training based on individual hormonal response patterns.[177] Four different workouts were compared to determine which would produce the maximal and minimal changes in salivary free T concentrations for each player. Training with the maximal T workouts (irrespective of their design) promoted increases in strength and body mass, whereas training with the minimal T workouts had no effect upon or reduced strength in most players. The prescription of resistance exercise and training programmes based on individual hormonal data is a novel area for research and should be pursued with *adhoc* studies with large athletic populations.

4.2 Chronic Modifications in Endogenous Hormones

4.2.1 Training Status

Resting or basal T and C concentrations are thought to represent the current state of muscle tissue and tissue homeostasis.[4,5] Modifications in basal T and/or C concentrations with resistance training are generally non-existent in untrained males.[14,15,57,142,147,149,161,178-180] There have been some reports of basal alterations in these hormones in studies on untrained males,[55,159,160,181] but these changes still tended to occur during the later training stages. In all of these studies, the relative improvements in performance were found to be much greater, and occurred much earlier, than any changes in body and/or muscle size. These data support the notion that the initial performance gains with training are largely due to neural adaptations, although endogenous T and C may still play a permissive role in this process.

Resistance training can modify the basal T and/or C concentrations of trained males.[15,150,182-194] This difference from untrained males may be attributed to adaptive changes in the neuromuscular and endocrine systems, combined with the greater training capacity (e.g. training intensity, total volume lifted) of trained men. Thus, whilst the training role of endogenous T and C may be permissive for untrained individuals, a dose-response role might exist for trained individuals. Indeed, these hormonal modifications have been used to explain the accompanying changes in speed, power or strength with training.[15,150,183-188,190-192,194] Nevertheless, some training studies on this population have noted performance changes without any modifications in basal T and/or C concentrations,[50,141,148,195-197] or *vice versa*.[52,158,198-201] Interpretation is limited by a lack of uniformity in study design, in particular the training status of subjects, as the term 'trained' can include a mixture of novice, intermediate or advanced lifters.

For elite athletes, alterations in basal hormones may be needed to induce further improvements in individual performance. Support

© 2011 Adis Data Information BV. All rights reserved.

comes from correlations between the individual changes in basal T and/or C concentrations and strength adaptation in elite strength athletes.[182,183,186,187,192,202] If these individual responses to training were related to genetic makeup, as discussed in section 4.1.4, then such data could be used to assess the trainability of some athletes. Research on power and strength athletes also indicates that basal changes in T and/or C concentrations may help to moderate, or maintain, performance during periods of detraining.[186,192,203] These findings corroborate a dose-dependent training role for endogenous T and C, especially for elite athletes, although full proof within the physiological range is still lacking. This model reconciles recent training studies that have proposed a permissive role for endogenous hormones in untrained individuals.[11,12,53]

4.2.2 Training Type

The type of training undertaken might also influence basal T. For example, sprinters had higher total T concentrations than soccer players, whilst soccer players in turn had higher total T than cross-country skiers.[143] Similar findings have been reported.[144,145] Conversely, endurance athletes often exhibit lower T concentrations than more explosive athletes and controls (table I). Thus, explosive training practices might lead to increases in T secretion whereas endurance training may lower T, possibly to meet the performance demands of each respective group. Training volume is one variable to consider with lower T identified in high mileage runners (vs moderate mileage runners and controls);[205] hence, a training threshold may well exist. The timing of assessment across the season can also influence the measured T values.[20] Alternatively, the differences identified between each athletic group could represent a genetic predisposition for a selected sport.

In general, basal C concentrations are no different between sporting and control groups (table I). There are still reports of greater basal

Table I. Summary of the basal testosterone (T) and cortisol (C) concentrations of different sporting groups and controls

Study	No. of subjects	Sessions	Samples[a]	Results
Arce et al.[204]	28 ER, BB and controls	1	1	Total T, free T: ER and BB < controls
De Souza et al.[205]	30 HER, MER and controls	1	1	Total T, free T: HER < MER and controls Total C, urinary free C: no group differences
Wittert et al.[206]	12 ETA and controls	1	22 (blood) 1 (urine)	Total C, urinary free C: no group differences
Lucía et al.[20]	40 ER, RC, TRI and controls	3	1	Total T: ER, RC and TRI < controls[b] Free T, total C: no group differences
Grasso et al.[207]	25 SP and controls	1	1	Total T: SP < controls Total C: no group differences
Hackney et al.[208]	88 ETA and controls	1	1	Total T, free T: ETA < controls Total C: no group differences
Maïmoun et al.[209]	48 ETA and controls	1	1	Total T: ETA < controls Total C: no group differences
Izquierdo et al.[210]	41 WL, RC and controls	1	1	Total T, free T: RC < WL and controls Total C, T : C: no group differences
Bambaeichi and Rahnama[145]	45 SR, ER and controls	1	1	Total T: ER < SR
Timon et al.[211]	30 ETA and controls	1	1	Urinary free T: ET < controls Urinary free C: no group differences
Bonifazi et al.[212]	26 SW, LTR and controls	1	1	Total C: no group differences

a All hormones measured in blood unless otherwise stated.

b First session only.

BB = bodybuilders; **ER** = endurance runners; **ETA** = endurance trained athletes; **HER** = high mileage endurance runners; **LTR** = low trained runners; **MER** = moderate mileage endurance runners; **RC** = road cyclists; **SP** = soccer players; **SR** = sprint runners; **SW** = swimmers; **TRI** = triathletes; **WL** = weightlifters.

© 2011 Adis Data Information BV. All rights reserved.

levels of adrenocorticotropic hormone (which stimulates C production) in endurance athletes than controls,[206,213] thereby suggesting a reduction in adrenal sensitivity with training. The acrophase and early morning increase in adrenocorticotropic hormone and total C also occurred earlier in endurance athletes than controls.[206] Together, these cross-sectional data on athletes and non-athletes highlight the potential of different training methods for regulating T secretion and the adrenal pathway that controls C secretion. One must remain cognizant of the limitations of cross-sectional research, such as the number of sessions monitored and the frequency and timing of sample collection, and the assumption that the observed results were due to training alone.

It appears from literature that the short-term effects of T and C may predominate in elite athletes, as others have suggested.[7,139,144] This supposition is based on the assumption that these endogenous hormones can moderate the performance/training outcomes and that muscle growth is either limited or non-existent in this athletic group. We acknowledge that the long-term T and C effects might still play a supporting role by maintaining lean body mass in some elite populations. Regardless, the exact contribution of these steroids to the training process is still unclear, because most endocrine studies have not taken accurate measurements of muscle size (e.g. muscle fibre cross-sectional area) and neural activity (e.g. electromyography, V-wave, Hoffmann reflex) to identify the training mechanisms involved. There is also a paucity of experimental studies on elite athletes that have specifically addressed the concept of dual steroid effects. More work is needed to delineate the specific hormonal mechanisms contributing to resistance training performance and adaptation.

4.3 Acute and Chronic Modifications in Steroid Receptors

4.3.1 Androgen Receptors

Steroid receptors mediate the hormonal effect by either up-regulating (i.e. increase in receptor content and/or binding sensitivity) or down-regulating (i.e. decrease in receptor content and/or binding sensitivity) in response to exercise and training.[4] If explosive training methods increased basal T production to support athlete performance, as indicated previously, then one may also expect the AR to be up-regulated. As supporting evidence, greater AR content was found in the trapezius muscle of power-lifters (vs controls), especially those taking anabolic steroids,[214] who themselves may exhibit a greater number and proportion of type II fibres. Transient adaptations are also evident, with resistance exercise promoting acute increases in both AR mRNA and/or protein content and T concentrations.[215-219] There are other reports of no changes, or decreases, in these AR measures after resistance exercise[220,221] and probably arising from their dynamic recovery patterns post-exercise.

It has been demonstrated that AR expression can vary between different muscles, muscle fibres and cells;[214,222-224] therefore, the responses evoked by different training methods may well show similar specificity. In rats,[225] resistance training down-regulated AR content in the soleus, which is predominantly a slow oxidative muscle, but up-regulated AR content in the extensor digitorum longus, which is primarily a fast glycolytic muscle. Conversely, endurance training increased AR content in the soleus, but did not change AR content in the extensor digitorum longus. Similar to basal T secretion, there does appear to be a loading threshold before acute exercise-induced changes in AR content may occur.[226,227]

4.3.2 Glucocorticoid Receptors

The GR are similarly responsive to exercise and training. In humans, a bout of resistance exercise acutely increased GR content in one study,[228] but not another,[221] although the workout tested and training background of subjects were different between these studies. Resistance training did provide an effective stimulus for increasing GR content in untrained males[229] and, in handball players, 10 weeks of training improved GR binding capacity.[230] Conversely, animal research have identified a reduction in GR content after endurance-type exercise,[231,232]

© 2011 Adis Data Information BV. All rights reserved.

with endurance training also reducing GR content in an intensity- and load-volume-dependent manner.[231,233,234] Cross-sectional evidence in humans supports these results.[207,212,233]

The specificity of the AR and GR responses to exercise and training is likely to support the biological effects at target tissue, by activating those signalling processes that contribute to the long-term steroid actions (i.e. gene transcription). However, because of compliance issues associated with muscle biopsies, most training studies in humans have used blood cells as indirect measures of muscle receptors; in particular, only the cell-bound type. Whether or not the membrane-bound receptors exhibit similar patterns of change with exercise and training, the time-course of these responses remains highly speculative.

4.4 Summary

Endogenous evidence supports the existence of short-term T and C effects on human performance within the resistance-training model. Several factors (e.g. workout design, nutrition, genetics, training status and type) can modify T and/or C concentrations and thereby potentially influence resistance-training performance and the adaptive outcomes. This novel pathway appears to be more prominent in athletes, possibly due to the training of the neuromuscular and endocrine systems. However, the exact contribution of these hormones to the training process is still unclear. A dose-response training role for endogenous T and C was confirmed, especially for elite athletes, but full proof is lacking. It is also clear that the steroid receptors (cell bound) mediate target tissue effects by adapting to exercise and training, but the response patterns of the membrane-bound receptors remain highly speculative.

5. Limitations and Challenges with Hormonal Research

This review has focused on the effects of T and C concentrations on the neuromuscular system. However, steroid concentrations still only reflect the secretory and clearance mechanisms.[235] Other important endocrine features include transport proteins, hormone-protein dissociation, hormone degradation rates and the intracrine and paracrine pathways.[235] The physiological meaning of the free and total steroids adds to this complexity, as does the different secretory mechanisms (e.g. rapid vs slow, blood vs brain). As outlined in section 3.2.4, the neuromuscular responses may also depend upon the actions and interactions of other steroid metabolites, neurotransmitters, peptides, amines and growth factors.

The importance of procedural issues (e.g. volume shifts, specimen collection, sleep) in endocrine research has been highlighted.[6,236,237] As an example, corrections should be made for plasma volume shifts when examining the T responses to exercise.[238,239] On the other hand, uncorrected hormonal values better represent what target tissue is actually exposed to, so the need to account for plasma volume depends on the research questions.[6] The collection of control data (or lack thereof) is another consideration because of the normal circadian variation in T and C secretion.[206,240-242] In fact, the exercise responses of T and/or C can also exhibit circadian variation.[243-245] The influence of other biological factors (e.g. age, sex, race, body composition) on these hormones and the implications for research analysis and interpretation have been discussed elsewhere.[236,237]

The analytical techniques employed can also influence research interpretation. Some of the endogenous evidence is based on correlations, which only imply cause and effect. However, correlations can help to identify relationships for further experimental testing[246] and may be useful when comparing group, sub-group and individual trends on athletes.[141,144,202] The method used for hormone testing (e.g. immunoassays, bioassays, chromatography) and their sensitivity, accuracy and repeatability for assessing hormones also requires attention.[236,237] A major limiting factor in the application of research into the sporting environment is the retrospective use of hormonal data because most testing methods require several hours for sample preparation and analysis. A real-time analysis system is implicit in being able to maximize the use of the hormonal information gained, for instance, by indicating a

© 2011 Adis Data Information BV. All rights reserved.

readiness to train or compete at a certain level, by determining the most appropriate warm-up strategy and/or by assisting with the design of workouts to promote optimal training gains.

Issues relating to research design are another consideration. Most of the exogenous evidence in this review comes from blinded, placebo-controlled, randomized trials whereas the endogenous evidence from athletes rarely meets this criterion. The strict control of training protocols is also difficult to achieve within the sporting environment and, notwithstanding, the confounding effects of other variables (e.g. competition, training periodization, psychology). Further bias comes from the selection criteria, smaller sample sizes and the lack of control data when undertaking research on elite athletes. Nevertheless, this research does represent the actual responses to, and inherent limitations within, elite sport and thus better represents the real-world setting.

From a training perspective, the interplay between the neuromuscular and endocrine systems presents a myriad of strategies for inducing performance gains with differential responses based on factors such as workout and training design, training status and type, age, sex, genetic predisposition and nutrition.[1,3,6,247] Resistance training can also induce other muscular (e.g. muscle fibre phenotype, fibre architecture), metabolic (e.g. substrate stores, enzyme activities) and/or neural changes (e.g. motor unit activation, firing rate, synchronization and reflex activity) to regulate the adaptive responses of the neuromuscular system.[3,21,22,247] Finally, many athletes often use resistance training to compliment other fitness- and sport-specific practices within a periodized programme,[2,172] so they may present multiple and sometimes conflicting systems of adaptation to produce the desired training effect.

6. Conclusion

Exogenous evidence confirms the existence of short-term T and C effects on the neuromuscular system and, through this, the expression of human performance. Whilst not conclusive, endogenous evidence from athletes provides some support regarding the short-term effects of these steroids on resistance training performance and the adaptive outcomes. Both forms of evidence also support a dose-dependent training role for T and C in moderating the performance capabilities and development of the neuromuscular system. For some elite athletes, changes (acute or basal) in endogenous T and C would seem to be one possible moderator of training performance and long-term adaptations. This athlete model reconciles a proposed permissive role for endogenous hormones in untrained individuals.

Acknowledgements

This review was written as part of the PhD thesis for Southern Cross University by the first author. Support for the preparation of this manuscript was provided by The New Zealand Institute for Plant & Food Research Limited, UK Sport and the Engineering and Physical Sciences Research Council of the UK. The authors have no conflicts of interest that are directly relevant to the content of this review.

References

1. Kraemer WJ, Adams K, Cafarelli E, et al. American College of Sports Medicine position stand: progression models in resistance training for healthy adults. Med Sci Sports Exerc 2002; 34 (2): 364-80
2. Kraemer WJ, Duncan ND, Volek JS. Resistance training and elite athletes: adaptations and program considerations. J Orthop Sport Phys 1998; 28 (2): 110-9
3. Folland JP, Williams AG. The adaptations to strength training: morphological and neurological contributions to increased strength. Sports Med 2007; 37 (2): 145-68
4. Kraemer WJ, Ratamess NA. Hormonal responses and adaptations to resistance exercise and training. Sports Med 2005; 35 (4): 339-61
5. Kraemer WJ, Mazzetti SA. Hormonal mechanisms related to the expression of muscular strength and power. In: Komi PV, editor. Strength and power in sport. Boston (MA): Blackwell Scientific Publishing, 2003: 73-95
6. Crewther B, Keogh J, Cronin J, et al. Possible stimuli for strength and power adaptation: acute hormonal responses. Sports Med 2006; 36 (3): 215-38
7. Viru A, Viru M. Preconditioning of the performance in power events by endogenous testosterone: in memory of Professor Carmelo Bosco. J Strength Cond Res 2005; 19 (1): 6-8
8. Makara GB, Haller J. Non-genomic effects of glucocorticoids in the neural system. Prog Neurobiol 2001; 65 (4): 367-90
9. Joëls M. Steroid hormones and excitability in the mammalian brain. Front Neuroendocrin 1997; 18 (1): 2-48
10. Falkenstein E, Tillman HC, Christ M, et al. Multiple actions of steroid hormones: a focus on rapid, nongenomic effects. Pharmacol Rev 2000; 52 (4): 513-55

© 2011 Adis Data Information BV. All rights reserved.

11. West DW, Burd NA, Tang JE, et al. Elevations in ostensibly anabolic hormones with resistance exercise enhance neither training-induced muscle hypertrophy nor strength of the elbow flexors. J Appl Physiol 2010; 108 (1): 60-7
12. West DW, Kujbida GW, Moore DR, et al. Resistance exercise-induced increases in putative anabolic hormones do not enhance muscle protein synthesis or intracellular signalling in young men. J Physiol 2009; 587 (21): 5239-47
13. Buresh R, Berg K, French J. The effect of resistive exercise rest interval on hormonal response, strength, and hypertrophy with training. J Strength Cond Res 2009; 23 (1): 62-71
14. Wilkinson SB, Tarnopolsky MA, Grant EJ, et al. Hypertrophy with unilateral resistance exercise occurs without increases in endogenous anabolic hormone concentration. Eur J Appl Physiol 2006; 98 (6): 546-55
15. Ahtiainen JP, Pakarinen A, Alén M, et al. Muscle hypertrophy, hormonal adaptations and strength development during strength training in strength-trained and untrained men. Eur J Appl Physiol 2003; 89 (6): 555-63
16. Tremblay MS, Copeland JL, Van Helder W. Effect of training status and exercise mode on endogenous steroid hormones in men. J Appl Physiol 2004; 96 (2): 531-9
17. Ahtiainen JP, Pakarinen A, Kraemer WJ, et al. Acute hormonal responses to heavy resistance exercise in strength athletes versus nonathletes. Can J Appl Physiol 2004; 29 (5): 527-43
18. Semmler JG, Nordstrom MA. Motor unit discharge and force tremor in skill- and strength-trained individuals. Exp Brain Res 1998; 119 (1): 27-38
19. Ryushi T, Häkkinen K, Kauhanen H, et al. Muscle fibre characteristics, muscle cross-sectional area and force production in strength athletes, physically active males and females. Scand J Sport Sci 1988; 10 (1): 7-15
20. Lucía A, Chicharro JL, Pérez M, et al. Reproductive function in male endurance athletes: sperm analysis and hormonal profile. J Appl Physiol 1996; 81 (6): 2627-36
21. Enoka RM. The motor system: multi-joint systems: chapter 7. In: Neuromechanics of human movement. 3rd ed. Champaign (IL): Human Kinetics, 2002: 313-58
22. Enoka RM. The motor system: single-joint system function: chapter 6. In: Neuromechanics of human movement. 3rd ed. Champaign (IL): Human Kinetics, 2002: 241-312
23. Brown M, Fisher JS, Hasser EM. Gonadectomy and reduced physical activity: effects on skeletal muscle. Arch Phys Med Rehab 2001; 82 (1): 93-7
24. Axell AM, MacLean HE, Plant DR, et al. Continuous administration prevents skeletal muscle atrophy and enhances resistance to fatigue in orchidectomized male mice. Am J Physiol Endocrinol Metab 2006; 291 (3): E506-16
25. Krotkiewski M, Kral JG, Karlsson J. Effects of castration and testosterone substitution on body composition and muscle metabolism in rats. Acta Physiol Scand 1980; 109 (3): 233-7
26. Ustunel I, Akkoyunlu G, Demir R. The effect of testosterone on gastrocnemius muscle fibres in growing and adult male and female rats: a histochemical, morphometric and ultrastructural study. Anat Histol Embryol 2003; 32 (2): 70-9
27. Gregory CM, Vandenborne K, Huang HF, et al. Effects of testosterone replacement therapy on skeletal muscle after spinal cord injury. Spinal Cord 2003; 41 (1): 23-8
28. Souccar C, Lapa AJ, Valle J. Influence of castration on the electrical excitability and contraction properties of the rat levator ani muscle. Exp Neurol 1982; 75 (3): 576-88
29. Lubischer JL, Arnold AP. Axotomy of developing rat spinal motoneurons: cell survival, soma size, muscle recovery, and the influence of testosterone. J Neurobiol 1995; 26 (2): 225-40
30. Czerwinski SM, Zak R, Kurowski TT, et al. Myosin heavy chain turnover and glucocorticoid deterrence by exercise in muscle. J Appl Physiol 1989; 67 (6): 2311-5
31. Falduto MT, Czerwinski SM, Hickson RC. Glucocorticoid-induced muscle atrophy prevention by exercise in fast-twitch fibers. J Appl Physiol 1990; 69 (3): 1058-62
32. Fimbel S, Abdelmalki A, Mayet MH, et al. Exercise training fails to prevent glucocorticoid-induced muscle alterations in young growing rats. Pflüg Arch Eur J Phy 1993; 424 (5-6): 369-76
33. Nakago K, Senda M, Touno M, et al. Influence of exercise on muscle fibers in rats with steroid myopathy. Acta Medica Okayama 1999; 53 (6): 265-70
34. Friedl KE, Dettori JR, Hannan CJ, et al. Comparison of the effects of high dose testosterone and 19-nortestosterone to a replacement dose of testosterone on strength and body composition in normal men. J Steroid Biochem 1991; 40 (4-6): 607-12
35. Bhasin S, Woodhouse L, Casaburi R, et al. Testosterone dose-response relationships in healthy young men. Am J Physiol-Endoc M 2001; 281 (6): E1172-81
36. Sinha-Hikim I, Artaza J, Woodhouse L, et al. Testosterone-induced increase in muscle size in healthy young men is associated with muscle fiber hypertrophy. Am J Physiol-Endoc M 2002; 283 (1): E154-64
37. Sinha-Hikim I, Cornford M, Gaytan H, et al. Effects of testosterone supplementation on skeletal muscle fibre hypertrophy and satellite cells in community-dwelling older men. J Clin Endocrinol Metab 2006; 91 (8): 3024-33
38. Woodhouse LJ, Reisz-Porszasz S, Javanbakht M, et al. Development of models to predict anabolic response to testosterone administration in healthy young men. Am J Physiol-Endoc M 2003; 284 (5): E1009-17
39. Bhasin S, Storer TW, Berman N, et al. Testosterone replacement increases fat-free mass and muscle size in hypogonadal men. J Clin Endocrinol Metab 1997; 82 (2): 407-13
40. Wang C, Swerdloff RS, Iranmanesh A, et al. Transdermal testosterone gel improves sexual function, mood, muscle strength, and body composition parameters in hypogonadal men. J Clin Endocrinol Metab 2000; 85 (8): 2839-53
41. McNurlan MA, Sandgren A, Hunter K, et al. Protein synthesis rates of skeletal muscle, lymphocytes, and albumin with stress hormone infusion in healthy man. Metabolism 1996; 45 (11): 1388-94
42. Brillon DJ, Zheng B, Campbell RG, et al. Effect of cortisol on energy expenditure and amino acid metabolism in humans. Am J Physiol 1995; 268 (3 Pt 1): E501-13
43. Ferrando AA, Stuart CA, Sheffield-Moore M, et al. Inactivity amplifies the catabolic response of skeletal muscle to cortisol. J Clin Endocrinol Metab 1999; 84 (10): 3515-21

© 2011 Adis Data Information BV. All rights reserved.

44. Fitts RH, Romatowski JG, Peters JR, et al. The deleterious effects of bed rest on human skeletal muscle fibers are exacerbated by hypercortisolemia and ameliorated by dietary supplementation. Am J Physiol Cell Physiol 2007; 293 (1): C313-20
45. Salehian B, Kejriwal K. Glucocorticoid-induced muscle atrophy: mechanisms and therapeutic strategies. Endocr Pract 1999; 5 (5): 277-81
46. Herbst KL, Bhasin S. Testosterone action on skeletal muscle. Curr Opin Clin Nutr Metab Care 2004; 7 (3): 271-7
47. Viru A, Viru M. Cortisol-essential adaptation hormone in exercise. Int J Sports Med 2004; 25 (6): 461-4
48. Bruce SA, Phillips SK, Woledge RC. Interpreting the relation between force and cross-sectional area in human muscle. Med Sci Sports Exerc 1997; 29 (5): 677-83
49. Storer TW, Magliano L, Woodhouse L, et al. Testosterone dose-dependently increases maximal voluntary strength and leg power, but does not affect fatigability or specific tension. J Clin Endocrinol Metab 2003; 88 (4): 1478-85
50. Bhasin S, Storer TW, Berman N, et al. The effects of supraphysiologic doses of testosterone on muscle size and strength in normal men. N Engl J Med 1996; 335 (1): 1-7
51. Rogerson S, Weatherby RP, Deakin GB, et al. The effect of short-term use of testosterone enanthate on muscular strength and power in healthy young men. J Strength Cond Res 2007; 21 (2): 354-62
52. Blazevich AJ, Giorgi A. Effect of testosterone administration and weight training on muscle architecture. Med Sci Sports Exerc 2001; 33 (10): 1688-93
53. Kvorning T, Andersen M, Brixen K, et al. Suppression of endogenous testosterone production attenuates the response to strength training: a randomized, placebo-controlled, and blinded intervention study. Am J Physiol Endocrinol Metab 2006; 291 (6): E1325-32
54. Czerwinski SM, Kurowski TG, O'Neill TM, et al. Initiating regular exercise protects against muscle atrophy from glucocorticoids. J Appl Physiol 1987; 63 (4): 1504-10
55. Staron RS, Karapondo DL, Kraemer WJ, et al. Skeletal muscle adaptations during early phase of heavy-resistance training in men and women. J Appl Physiol 1994; 76 (3): 1247-55
56. Moritani MA, deVries HA. Neural factors versus hypertrophy in the time course of muscle strength gain. Am J Phys Med 1979; 58 (3): 115-30
57. Hickson RC, Hidaka K, Foster C, et al. Successive time courses of strength development and steroid hormone responses to heavy-resistance training. J Appl Physiol 1994; 76 (2): 663-70
58. Komi P. Training of muscle strength and power: interaction of neuromotoric, hypertrophic, and mechanical factors. Int J Sports Med 1986; 7 (1 Suppl.): 10-5
59. Abe T, DeHoyos D, Pollock ML, et al. Time course for strength and muscle thickness changes following upper and lower body resistance training in men and women. Eur J Appl Physiol 2000; 81 (3): 174-80
60. Kurz EM, Sengelaub DR, Arnold AP. Androgens regulate the dendritic length of mammalian motoneurons in adulthood. Science 1986; 232 (4748): 395-8
61. Fargo KN, Sengelaub DR. Testosterone manipulation protects motoneurons from dendritic atrophy after contralateral motoneuron depletion. J Comp Neurol 2004; 469 (1): 96-106
62. Matsumoto A, Micevych PE, Arnold AP. Androgen regulates synaptic input to motoneurons of the adult rat spinal cord. J Neurosci 1988; 8 (11): 4168-76
63. Breedlove SM, Arnold AP. Sexually dimorphic motor nucleus in the rat lumbar spinal cord: response to adult hormone manipulation, absence in androgen-insensitive rats. Brain Res 1981; 225 (2): 297-307
64. Siegford JM, Ulibarri C. Increase in motoneurons in the spinal nucleus of the bulbocavernosus of prepubertally castrated male Mongolian gerbils following delayed treatment with testosterone. J Comp Neurol 2004; 473 (4): 485-95
65. Manolides LS, Baloyannis SJ. Influence of hydrocortisone, progesterone and testosterone on dendritic growth in vitro. Acta Otolaryngol 1984; 97 (5-6): 509-22
66. Sapolsky RM, Uno H, Rebert CS, et al. Hippocampal damage associated with prolonged glucocorticoid exposure in primates. J Neurosci 1990; 10 (9): 2897-902
67. Askanas V, McFerrin J, Park-Matsumoto YC, et al. Glucocorticoid increases acetylcholinesterase and organization of the postsynaptic membrane in innervated cultured human muscle. Exp Neurol 1992; 115 (3): 368-75
68. Heinlein CA, Chang C. The roles of androgen receptors and androgen-binding proteins in nongenomic androgen actions. Mol Endocrinol 2002; 16 (10): 2181-7
69. Han JZ, Lin W, Lou SJ, et al. A rapid, nongenomic action of glucocorticoids in rat B103 neuroblastoma cells. Biochim Biophys Acta 2002; 1591 (1-3): 21-7
70. Han JZ, Lin W, Chen YZ. Inhibition of ATP-induced calcium influx in HT4 cells by glucocorticoids: involvement of protein kinase A. Acta Pharm Sinic 2005; 26 (2): 199-204
71. Passaquin AC, Lhote P, Rüegg UT. Calcium influx inhibition by steroids and analogs in C2C12 skeletal muscle cells. Brit J Pharmacol 1998; 124 (8): 1751-9
72. Jaimovich E, Espinosa A. Possible link of different slow calcium signals generated by membrane potential and hormones to differential gene expression in cultured muscle cells. Biol Res 2004; 37 (4): 625-33
73. Estrada M, Liberona JL, Miranda M, et al. Aldosterone- and testosterone-mediated intracellular calcium response in skeletal muscle cell cultures. Am J Physiol Endocrinol Metab 2000; 279 (1): E132-9
74. Estrada M, Espinosa A, Müller M, et al. Testosterone stimulates intracellular calcium release and mitogen-activated protein kinases via a G protein-coupled receptor in skeletal muscle cells. Endocrinology 2003; 144 (8): 3586-97
75. Sunny F, Oommen OV. Rapid action of glucocorticoids on branchial ATPase activity in *Oreochromis mossambicus*: an *in vivo* and *in vitro* study. Comp Biochem Physiol B Biochem Mol Biol 2001; 130 (3): 323-30
76. Ding AQ, Stallone JN. Testosterone-induced relaxation of rat aorta is androgen structure specific and involves K+ channel activation. J Appl Physiol 2001; 91 (6): 2742-50
77. Bai CX, Kurokawa J, Tamagawa M, et al. Nontranscriptional regulation of cardiac repolarization currents by testosterone. Circulation 2005; 112 (12): 1701-10

© 2011 Adis Data Information BV. All rights reserved.

78. Nguyen TV, Yao M, Pike CJ. Androgens activate mitogen-activated protein kinase signalling: role in neuroprotection. J Neurochem 2005; 94 (6): 1639-51
79. Hatanaka Y, Mukai H, Mitsuhashi K, et al. Androgen rapidly increases dendritic thorns of CA3 neurons in male rat hippocampus. Biochem Biophys Res Commun 2009; 381 (4): 728-32
80. Jansen HT, Popiela CL, Jackson GL, et al. A re-evaluation of the effects of gonadal steroids on neuronal activity in the male rat. Brain Res Bull 1993; 31 (1-2): 217-23
81. Smith MD, Jones LS, Wilson MA. Sex differences in hippocampal slice excitability: role of testosterone. Neuroscience 2002; 109 (3): 517-30
82. Chen YZ, Hua SY, Wang CA, et al. An electrophysiological study on the membrane receptor-mediated action of glucocorticoids in mammalian neurons. Neuroendocrinol 1991; 53 (1 Suppl.): 25S-30S
83. Kasai M, Yamashita H. Inhibition by cortisol of neurons in the paraventricular nucleus of the hypothalamus in adrenalectomized rats: an *in vitro* study. Neurosci Lett 1988; 91 (1): 59-64
84. Zaki A, Barrett-Jolley R. Rapid neuromodulation by cortisol in the rat paraventricular nucleus: an *in vitro* study. Brit J Pharmacol 2002; 137 (1): 87-97
85. James PJ, Nyby JG. Testosterone rapidly affects the expression of copulatory behavior in house mice (Mus musculus). Physiol Behav 2002; 75 (3): 287-94
86. Aikey JL, Nyby JG, Anmuth DM, et al. Testosterone rapidly reduces anxiety in male house mice (Mus musculus). Horm Behav 2002; 42 (4): 448-60
87. Schiml PA, Rissman EF. Cortisol facilitates induction of sexual behavior in the female musk shrew (Suncus murinus). Behav Neurosci 1999; 113 (1): 166-75
88. Remage-Healey L, Bass AH. From social behavior to neural circuitry: steroid hormones rapidly modulate advertisement calling via a vocal pattern generator. Horm Behav 2006; 50 (3): 432-41
89. Hartgens F, Kuipers H. Effects of androgenic-anabolic steroids in athletes. Sports Med 2004; 34 (8): 513-54
90. O'Connor DB, Archer J, Hair WM, et al. Activational effects of testosterone on cognitive function in men. Neuropsychobiologia 2001; 39 (13): 1385-94
91. Aleman A, Bronk E, Kessels RPC, et al. A single administration of testosterone improves visuospatial ability in young women. Psychoneuroendocrino 2004; 29 (5): 612-7
92. Hsu FC, Garside MJ, Massey AE, et al. Effects of a single dose of cortisol on the neural correlates of episodic memory and error processing in healthy volunteers. Psychopharmacology 2003; 167 (4): 431-42
93. Buss C, Wolf OT, Witt J, et al. Autobiographic memory impairment following acute cortisol administration. Psychoneuroendocrino 2004; 29 (8): 1093-6
94. Hermans EJ, Putman P, Baas JM, et al. A single administration of testosterone reduces fear-potentiated startle in humans. Biol Psychiat 2006; 59 (9): 872-4
95. Putman P, Antypa N, Crysovergi P, et al. Exogenous cortisol acutely influences motivated decision making in healthy young men. Psychopharmacology 2010; 208 (2): 257-63
96. van Honk J, Peper JS, Schutter DJ. Testosterone reduces unconscious fear but not consciously experienced anxiety: implications for the disorders of fear and anxiety. Biol Psychiat 2005; 58 (3): 218-25
97. Lacreuse A, Chiavetta MR, Shirai AC, et al. Effects of testosterone on cognition in young adult male rhesus monkeys. Physiol Behav 2009; 98 (5): 524-31
98. Lyons DM, Lopez JM, Yang C, et al. Stress-level cortisol treatment impairs inhibitory control of behavior in monkeys. J Neurosci 2000; 20 (20): 7816-21
99. Sale MV, Ridding MC, Nordstorm MA. Cortisol inhibits neuroplasticity induction in human motor cortex. J Neurosci 2008; 28 (33): 8285-93
100. Bonifazi M, Ginanneschi F, della Volpe R, et al. Effects of gonadal steroids on the input-output relationship of the corticospinal pathway in humans. Brain Res 2004; 1011 (2): 187-94
101. Banks WA, Morley JE, Niehoff ML, et al. Delivery of testosterone to the brain by intranasal administration: comparison to intravenous testosterone. J Drug Target 2009; 17 (2): 91-7
102. James P, Rivier C, Lee S. Presence of corticotrophin-releasing factor and/or tyrosine hydroxylase in cells of a neural brain-testicular pathway that are labelled by a transganglionic tracer. J Neuroendocrinol 2008; 20 (2): 173-81
103. Selvage DJ, Rivier C. Importance of the paraventricular nucleus of the hypothalamus as a component of a neural pathway between the brain and the testes that modulates testosterone secretion independently of the pituitary. Endocrinology 2003; 144 (2): 594-8
104. Selvage DJ, Lee SY, Parsons LH, et al. A hypothalamic-testicular neural pathway is influenced by brain catecholamines, but not testicular blood flow. Endocrinology 2004; 145 (4): 1750-9
105. Stromberg J, Backstrom T, Lundgren P. Rapid non-genomic effect of glucocorticoid metabolites and neurosteroids on the gamma-aminobutyric acid-A receptor. Eur J Neurosci 2005; 21 (8): 2083-8
106. Weill-Engerer S, David JP, Sazdovitch V, et al. Neurosteroid quantification in human brain regions: comparison between Alzheimer's and nondemented patients. J Clin Endocrinol Metab 2002; 87 (11): 5138-43
107. Ebner MJ, Corol DI, Havlíková H, et al. Identification of neuroactive steroids and their precursors and metabolites in adult male rat brain. Endocrinology 2006; 147 (1): 179-90
108. Ishii H, Tsurugizawa T, Ogiue-Ikeda M, et al. Local production of sex hormones and their modulation of hippocampal synaptic plasticity. Neuroscientist 2007; 13 (4): 323-34
109. Summers CH, Larson ET, Ronan PJ, et al. Serotonergic responses to corticosterone and testosterone in the limbic system. Gen Comp Endocrinol 2000; 117 (1): 151-9
110. Berchtold MW, Brinkmeier H, Muntener M. Calcium ion in skeletal muscle: its crucial role for muscle function, plasticity, and disease. Physiol Rev 2000; 80 (3): 1215-65
111. Anttila K, Mänttäri S, Järvilehto M. Testosterone and Ca2+ regulation in skeletal muscle. Int J Sports Med 2008; 29 (10): 795-802

© 2011 Adis Data Information BV. All rights reserved.

112. Braun S, Sarkozi E, McFerrin J, et al. Hydrocortisone influences voltage-dependent L-type calcium channels in cultured human skeletal muscle. J Neurosci Res 1995; 41 (6): 727-33
113. Clausen T. Na^+-K^+ pump regulation and skeletal muscle contractility. Physiol Rev 2003; 83 (4): 1269-323
114. Dlouhá H, Vyskočil F. The effect of cortisol on the excitability of the rat muscle fibre membrane and neuromuscular transmission. Physiol Bohemoslov 1979; 28 (6): 485-94
115. Sachs BD, Leipheimer RE. Rapid effect of testosterone on striated muscle activity in rats. Neuroendocrinol 1988; 48 (5): 453-8
116. Blanco CE, Zhan W, Fang Y, et al. Exogenous testosterone treatment decreases diaphragm neuromuscular transmission failure in male rats. J Appl Physiol 2001; 90 (3): 850-6
117. Stewart PM. The adrenal cortex. In: Larsen RP, Kronoberg HM, Melmed S, et al., editors. Williams textbook of endocrinology. 10th ed. Philadelphia (PA): W.B. Saunders, 2002: 491-551
118. Gorostiaga EM, Czerwinski SM, Hickson RC. Acute glucocorticoid effects on glycogen utilization, O_2 uptake, and endurance. J Appl Physiol 1988; 64 (3): 1098-106
119. Dodt C, Keyser B, Molle M, et al. Acute suppression of muscle sympathetic nerve activity by hydrocortisone in humans. Hypertension 2000; 35 (3): 758-63
120. Vozarova B, Weyer C, Snitker S, et al. Effect of cortisol on muscle sympathetic nerve activity in Pima Indians and Caucasians. J Clin Endocrinol Metab 2003; 88 (7): 3218-26
121. Tsai LW, Sapolsky RM. Rapid stimulatory effects of testosterone upon myotubule metabolism and sugar transport, as assessed by silicon microphysiometry. Aggressive Behav 1996; 22 (2): 357-64
122. Holmang A, Bjorntorp P. The effects of testosterone on insulin sensitivity in male rats. Acta Physiol Scand 1992; 146 (4): 505-10
123. Holmang A, Larsson BM, Brzezinska Z, et al. Effects of short-term testosterone exposure on insulin sensitivity of muscles in female rats. Am J Physiol 1992; 262: E851-5
124. Braun B, Gerson L, Hagobian T, et al. No effect of short-term testosterone manipulation on exercise substrate metabolism in men. J Appl Physiol 2005; 99 (5): 1930-7
125. Ramamani A, Aruldhas MM, Govindarajulu P. Impact of testosterone and oestradiol on region specificity of skeletal muscle-ATP, creatine phosphokinase and myokinase in male and female Wistar rats. Acta Physiol Scand 1999; 166 (2): 91-7
126. Ramamani A, Aruldhas MM, Govindarajulu P. Differential response of rat skeletal muscle glycogen metabolism to testosterone and estradiol. Can J Physiol Pharmacol 1999; 77 (4): 300-4
127. Pastoris O, Dossena M, Fulle D, et al. Action of testosterone on some biochemical parameters related to the energy metabolism of the skeletal muscle. Arch Int Pharmacodyn Ther 1983; 263 (1): 129-38
128. van Breda E, Keizer HA, Geurten P, et al. Modulation of glycogen metabolism of rat skeletal muscles by endurance training and testosterone treatment. Pflügers Arch 1993; 424 (3-4): 294-300
129. Enoka RM. Adaptability of the motor system: acute adjustments: chapter 8. In: Neuromechanics of human movement. 3rd ed. Champaign (IL): Human Kinetics, 2002: 363-96
130. McCaulley GO, McBride JM, Cormie P, et al. Acute hormonal and neuromuscular responses to hypertrophy, strength and power type resistance exercise. Eur J Appl Physiol 2009; 105 (5): 695-704
131. Crewther B, Cronin J, Keogh J, et al. The salivary testosterone and cortisol response to three loading schemes. J Strength Cond Res 2008; 22 (1): 250-5
132. Smilios I, Pilianidis T, Karamouzis M, et al. Hormonal responses after various resistance exercise protocols. Med Sci Sports Exerc 2003; 35 (4): 644-54
133. Häkkinen K, Pakarinen A. Acute hormonal responses to two different fatiguing heavy-resistance protocols in male athletes. J Appl Physiol 1993; 74 (2): 882-7
134. Kraemer WJ, Gordon SE, Fleck SJ, et al. Endogenous anabolic hormonal and growth factor responses to heavy resistance exercise in males and females. Int J Sports Med 1991; 12 (2): 228-35
135. Linnamo V, Pakarinen A, Komi PV, et al. Acute hormonal responses to submaximal and maximal heavy resistance and explosive exercises in men and women. J Strength Cond Res 2005; 19 (3): 566-71
136. Crewther B, Cronin J, Keogh J. Possible stimuli for strength and power adaptation: acute mechanical responses. Sports Med 2005; 35 (11): 967-89
137. Crewther BT, Cronin J, Keogh JWL. The contribution of volume, technique, and load to single-repetition and total-repetition kinematics and kinetics in response to three loading schemes. J Strength Cond Res 2008; 22 (6): 1908-15
138. Keogh J, Wilson G, Weatherby R. A cross-sectional comparison of different resistance training techniques in the bench press. J Strength Cond Res 1999; 13 (3): 247-58
139. Crewther BT, Cook CJ, Lowe TE, et al. The effects of short cycle sprints on power, strength and salivary hormones in elite rugby players. J Strength Cond Res. Epub 2010 Jan 21
140. Obmiński Z, Borkowski L, Ladyga M, et al. Concentrations of cortisol, testosterone and lactate, and power output in repeated, supramaximal exercise in elite fencers. Biol Sport 1998; 15 (1): 19-24
141. Crewther BT, Lowe T, Weatherby RP, et al. Prior sprint cycling did not enhance training adaptation, but resting salivary hormones were related to workout power and strength. Eur J Appl Physiol 2009; 105 (6): 919-27
142. Hansen S, Kvorning T, Kjaer M, et al. The effect of short-term strength training on human skeletal muscle: the importance of physiologically elevated hormone levels. Scand J Med Sci Sport 2001; 11 (6): 347-54
143. Bosco C, Viru A. Testosterone and cortisol levels in blood of male sprinters, soccer players and cross-country skiers. Biol Sport 1998; 15 (1): 3-8
144. Cardinale M, Stone MH. Is testosterone influencing explosive performance? J Strength Cond Res 2006; 20 (1): 103-7
145. Bambaeichi E, Rahnama N. Comparison of testosterone, progesterone and oestradiol concentrations between

© 2011 Adis Data Information BV. All rights reserved.

sprint runners, endurance runners and untrained males. J Sports Sci 2005; 23 (2): 187-8

146. Kraemer WJ, Hatfield DL, Spiering BA, et al. Effects of a multi-nutrient supplement on exercise performance and hormonal responses to resistance exercise. Eur J Appl Physiol 2007; 101 (5): 637-46
147. Bird SP, Tarpenning KM, Marino FE. Independent and combined effects of liquid carbohydrate/essential amino acid ingestion on hormonal and muscular adaptations following resistance training in untrained men. Eur J Appl Physiol 2006; 97 (2): 225-38
148. Tarpenning KM, Wiswell RA, Hawkins SA, et al. Influence of weight training exercise and modification of hormonal response on skeletal muscle growth. J Sci Med Sport 2001; 4 (4): 431-46
149. Kraemer WJ, Hatfield DL, Volek JS, et al. Effects of amino acids supplement on physiological adaptations to resistance training. Med Sci Sports Exerc 2009; 41 (5): 1111-21
150. Kraemer WJ, Ratamess NA, Volek JS, et al. The effects of amino acid supplementation on hormonal responses to resistance training overreaching. Metabolism 2006; 55 (3): 282-91
151. Hulmi JJ, Volek JS, Selänne H, et al. Protein ingestion prior to strength exercise affects blood hormones and metabolism. Med Sci Sports Exerc 2005; 37 (11): 1990-7
152. Kraemer W, Volek J, Bush J, et al. Hormonal responses to consecutive days of heavy-resistance exercise with or without nutritional supplementation. J Appl Physiol 1998; 85 (4): 1544-55
153. Thyfault JP, Carper MJ, Richmond S, et al. Effects of liquid carbohydrate ingestion on markers of anabolism following high-intensity resistance exercise. J Strength Cond Res 2004; 18 (1): 174-9
154. Ratamess NA, Hoffman JR, Ross R, et al. Effects of an amino acid/creatine energy supplement on the acute hormonal response to resistance exercise. Int J Sport Nutr Exerc Metab 2007; 17 (6): 608-23
155. Hoffman JR, Ratamess NA, Ross R, et al. Effect of a pre-exercise energy supplement on the acute hormonal response to resistance exercise. J Strength Cond Res 2008; 22 (3): 874-82
156. Williams AG, Ismail AN, Sharma A, et al. Effects of resistance exercise volume and nutritional supplementation on anabolic and catabolic hormones. Eur J Appl Physiol 2002; 86 (4): 315-21
157. Baty JJ, Hwang H, Ding Z, et al. The effect of a carbohydrate and protein supplement on resistance exercise performance, hormonal response, and muscle damage. J Strength Cond Res 2007; 21 (2): 321-9
158. Fry AC, Kraemer WJ, Stone MH, et al. Endocrine and performance responses to high volume training and amino acid supplementation in elite junior weightlifters. Int J Sport Nutr 1993; 3 (3): 306-22
159. Kraemer WJ, Häkkinen K, Newton RU, et al. Effects of heavy-resistance training on hormonal response patterns in younger vs. older men. J Appl Physiol 1999; 87 (3): 982-92
160. Kraemer WJ, Staron R, Hagerman F, et al. The effects of short-term resistance training on endocrine function in men and women. Eur J Appl Physiol Occup Physiol 1998; 78 (1): 69-76
161. Izquierdo M, Ibanez J, Calbert JA, et al. Cytokine and hormone responses to resistance training. Eur J Appl Physiol 2009; 107 (4): 397-409
162. Kraemer WJ, Fry AC, Warren BJ, et al. Acute hormonal responses in elite junior weightlifters. Int J Sports Med 1992; 13 (2): 103-9
163. Bosco C, Colli R, Bonomi R, et al. Monitoring strength training: neuromuscular and hormonal profile. Med Sci Sports Exerc 2000; 32 (1): 202-8
164. Słowińska-Lisowska M, Witkowski K. The influence of exercise on the functioning of the pituitary-gonadal axis in physically active older and younger men. Aging Male 2001; 4 (3): 145-50
165. Bosco C, Tihanyi J, Rivalta L, et al. Hormonal responses in strenuous jumping effort. Jap J Physiol 1996; 46 (1): 93-8
166. Jürimäe J, Jürimäe T. Responses of blood hormones to the maximal rowing ergometer test in college rowers. J Sports Med Phys Fit 2001; 41 (1): 73-7
167. Słowińska-Lisowska M, Majda J. Hormone plasma levels from pituitary-gonadal axis in performance athletes after the 400 m run. J Sports Med Phys Fit 2002; 42 (2): 243-9
168. Boone JB, Lambert CP, Flynn MG, et al. Resistance exercise effects on plasma cortisol, testosterone and creatine kinase activity in anabolic-androgenic steroid users. Int J Sports Med 1990; 11 (4): 293-7
169. Tsopanakis A, Stalikas A, Sgouraki E, et al. Stress adaptation in athletes: relation of lipoprotein levels to hormonal response. Pharmacol Biochem Behav 1994; 48 (2): 377-82
170. Tesch P, Karlsson J. Muscle fiber types and size in trained and untrained muscles of elite athletes. J Appl Physiol 1985; 59 (6): 1716-20
171. Tesch P, Thorsson A, Essen-Gustavsson B. Enzyme activities of FT and ST muscle fibres in heavy-resistance trained athletes. J Appl Physiol 1989; 67 (1): 83-7
172. Smith DJ. A framework for understanding the training process leading to elite performance. Sports Med 2003; 33 (15): 1103-26
173. Beaven MC, Gill ND, Cook CJ. Salivary testosterone and cortisol responses in professional rugby players after four resistance exercise protocols. J Strength Cond Res 2008; 22 (2): 426-32
174. Jensen J, Oftebro H, Breigan B, et al. Comparison of changes in testosterone concentrations after strength and endurance exercise in well trained men. Eur J Appl Physiol Occup Physiol 1991; 63 (6): 467-71
175. Viru A, Karelson K, Smirnova T. Stability and variability in hormonal responses to prolonged exercise. Int J Sports Med 1992; 13 (3): 230-5
176. Di Luigi L, Guidetti L, Baldari C, et al. Heredity and pituitary response to exercise-related stress in trained men. Int J Sports Med 2003; 24 (8): 551-8
177. Beaven MC, Cook CJ, Gill ND. Significant strength gains observed in rugby players after specific resistance exercise protocols based on individual salivary testosterone responses. J Strength Cond Res 2008; 22 (2): 419-25
178. Craig BW, Brown R, Everhart J. Effects of progressive resistance training on growth hormone and testosterone

© 2011 Adis Data Information BV. All rights reserved.

levels in young and elderly subjects. Mech Ageing Dev 1989; 49 (2): 159-69

179. Bell G, Syrotuik D, Martin TP, et al. Effect of concurrent strength and endurance training on skeletal muscle properties and hormone concentrations in humans. Eur J Appl Physiol 2000; 81 (5): 418-27
180. Kvorning T, Bagger M, Caserotti P, et al. Effects of vibration and resistance training on neuromuscular and hormonal measures. Eur J Appl Physiol 2006; 96 (5): 615-25
181. Sedliak M, Finni T, Cheng S, et al. Effect of time-of-day-specific strength training on serum hormone concentrations and isometric strength in men. Chronobiol Int 2007; 24 (6): 1159-77
182. Häkkinen K, Pakarinen A, Alén M, et al. Neuromuscular and hormonal adaptations in athletes to strength training in two years. J Appl Physiol 1988; 65 (6): 2406-12
183. Fry AC, Kraemer WJ, Stone MH, et al. Relationships between serum testosterone, cortisol, and weightlifting performance. J Strength Cond Res 2000; 14 (3): 338-43
184. Fry AC, Kraemer WJ, Ramsey LT. Pituitary-adrenal-gonadal responses to high-intensity resistance exercise overtraining. J Appl Physiol 1998; 85 (6): 2352-9
185. Häkkinen K, Pakarinen A. Serum hormones in male strength athletes during intensive short term strength training. Eur J Appl Physiol Occ Physiol 1991; 63 (3-4): 194-9
186. Alén M, Pakarinen A, Häkkinen K, et al. Responses of serum androgenic-anabolic and catabolic hormones to prolonged strength training. Int J Sports Med 1988; 9 (3): 229-33
187. Raastad T, Glomsheller T, Bjøro T, et al. Changes in human skeletal muscle contractility and hormone status during 2 weeks of heavy strength training. Eur J Appl Physiol 2001; 84 (1-2): 54-63
188. Potteiger JA, Judge LW, Cerny JA, et al. Effects of altering training volume and intensity on body mass, performance, and hormonal concentrations in weight-event athletes. J Strength Cond Res 1995; 9 (1): 55-8
189. Bell G, Syrotuik D, Socha T, et al. Effect of strength training and concurrent strength and endurance training on strength, testosterone, and cortisol. J Strength Cond Res 1997; 11 (1): 57-64
190. Moore CA, Fry AC. Nonfunctional overreaching during off-season training for skill position players in collegiate American football. J Strength Cond Res 2007; 21 (3): 793-800
191. Izquierdo M, Ibañez J, Gonzalez-Badillo J, et al. Differential effects of strength training leading to failure versus not to failure on hormonal responses, strength, and muscle power gains. J Appl Physiol 2006; 100 (5): 1647-56
192. Häkkinen K, Pakarinen A, Alén M, et al. Serum hormones during prolonged training of neuromuscular performance. Eur J Appl Physiol Occup Physiol 1985; 53 (4): 287-93
193. McCall GE, Byrnes WC, Fleck SJ, et al. Acute and chronic hormonal responses to resistance training designed to promote muscle hypertrophy. Can J Appl Physiol 1999; 24 (1): 96-107
194. Hartman MJ, Clark B, Bembens DA, et al. Comparisons between twice-daily and once-daily training sessions in male weight lifters. Int J Sports Physiol Perform 2007; 2 (2): 159-69
195. Ostrowski JK, Wilson J, Weatherby R, et al. The effect of weight training volume on hormonal output and muscular size and function. J Strength Cond Res 1997; 11 (1): 148-54
196. Izquierdo M, Ibañez J, Gonzalez-Badillo JJ, et al. Detraining and tapering effects on hormonal responses and strength performance. J Strength Cond Res 2007; 21 (3): 768-75
197. Ahtiainen JP, Pakarinen A, Alén M, et al. Short vs. long rest period between the sets in hypertrophic resistance training: influence on muscle strength, size, and hormonal adaptations in trained men. J Strength Cond Res 2005; 19 (3): 572-82
198. Fry AC, Kraemer WJ, Stone MH, et al. Endocrine responses to overreaching before and after 1 year of weightlifting. Can J Appl Physiol 1994; 19 (4): 400-10
199. Häkkinen K. Neuromuscular and hormonal adaptations during strength and power training. J Sports Med Phys Fit 1989; 29 (1): 9-26
200. Häkkinen K, Pakarinen A, Alén M, et al. Daily hormonal and neuromuscular responses to intensive strength training in 1 week. Int J Sports Med 1988; 9 (6): 422-8
201. Häkkinen K, Pakarinen A, Alén M, et al. Relationships between training volume, physical performance capacity, and serum hormone concentrations during prolonged training in elite weight lifters. Int J Sports Med 1987; 8 (1 Suppl.): 61S-5S
202. Crewther BT, Cook C. Relationships between salivary testosterone and cortisol concentrations and training performance in Olympic weightlifters. J Sports Med Phys Fitness 2010; 50 (3): 166-73
203. Hortobágyi T, Houmard JA, Stevenson JR, et al. The effects of detraining on power athletes. Med Sci Sports Exerc 1993; 25 (8): 929-35
204. Arce JC, De Souza MJ, Pescatello LS, et al. Subclinical alterations in hormone and semen profiles in athletes. Fertil Steril 1993; 59 (2): 398-404
205. De Souza MJ, Arce JC, Pescatello LS, et al. Gonadal hormones and semen quality in male runners. Int J Sports Med 1994; 15 (7): 383-91
206. Wittert GA, Livesey JH, Espiner EA, et al. Adaptation of the hypothalamopituitary adrenal axis to chronic exercise stress in humans. Med Sci Sports Exerc 1996; 28 (8): 1015-9
207. Grasso G, Lodi L, Lupo C, et al. Glucocorticoid receptors in human peripheral blood mononuclear cells in relation to age and to sport activity. Life Sci 1997; 61 (3): 301-8
208. Hackney AC, Fahrner CL, Gulledge TP. Basal reproductive hormonal profiles are altered in endurance trained men. J Sports Med Phys Fit 1998; 38 (2): 138-41
209. Maïmoun L, Lumbroso S, Manetta J, et al. Testosterone is significantly reduced in endurance athletes without impact on bone mineral density. Horm Res 2003; 59 (6): 285-92
210. Izquierdo M, Ibañez J, Häkkinen K, et al. Maximal strength and power, muscle mass, endurance and serum

© 2011 Adis Data Information BV. All rights reserved.

hormones in weightlifters and road cyclists. J Sports Sci 2004; 22 (5): 465-78

211. Timon R, Olcina G, Maynar M, et al. Evaluation of urinary steroid profile in highly trained cyclists. J Sports Med Phys Fit 2008; 48 (8): 530-4
212. Bonifazi M, Mencarelli M, Fedele V, et al. Glucocorticoid receptor mRNA expression in peripheral blood mononuclear cells in high trained compared to low trained athletes and untrained subjects. J Endocrinol Invest 2009; 32 (10): 816-20
213. Duclos M, Corcuff JB, Rashedi M, et al. Trained versus untrained men: different immediate post-exercise responses of pituitary adrenal axis. A preliminary study. Eur J Appl Physiol Occup Physiol 1997; 75 (4): 343-50
214. Kadi F, Bonnerud P, Eriksson A, et al. The expression of androgen receptors in human neck and limb muscles: effects of training and self-administration of androgenic-anabolic steroids. Histochem Cell Biol 2000; 113 (1): 25-9
215. Willoughby DS, Taylor L. Effects of sequential bouts of resistance exercise on androgen receptor expression. Med Sci Sports Exerc 2004; 36 (9): 1499-506
216. Spiering BA, Kraemer WJ, Vingren JL, et al. Elevated endogenous testosterone concentrations potentiate muscle androgen receptor responses to resistance exercise. J Steroid Biochem 2009; 114 (3-5): 195-9
217. Hulmi JJ, Ahtiainen JP, Selänne H, et al. Androgen receptors and testosterone in men: effects of protein ingestion, resistance exercise and fiber type. J Steroid Biochem 2008; 110 (1-2): 130-7
218. Bamman MM, Shipp JR, Jiang J, et al. Mechanical load increases muscle IGF-1 and androgen receptor mRNA concentrations in humans. Am J Physiol Endocrinol Metab 2001; 280 (3): E383-90
219. Kraemer WJ, Spiering BA, Volek JS, et al. Androgenic responses to resistance exercise: effects of feeding and L-carnitine. Med Sci Sports Exerc 2006; 38 (7): 1288-96
220. Ratamess NA, Kraemer WJ, Volek JS, et al. Androgen receptor content following heavy resistance exercise in men. J Steroid Biochem 2005; 93 (1): 35-42
221. Vingren JL, Kraemer WJ, Hatfield DL, et al. Effect of resistance exercise on muscle steroid receptor protein content in strength-trained men and women. Steroids 2009; 74 (13-14): 1033-9
222. Sinha-Hikim I, Taylor WE, Gonzalez-Cadavid NF, et al. Androgen receptor in human skeletal muscle and cultured muscle satellite cells: up-regulation by androgen treatment. J Clin Endocrinol Metab 2004; 89 (10): 5245-55
223. Johansen JA, Breedlove SM, Jordan CL. Androgen receptor expression in the levator ani muscle of male mice. J Neuroendocrinol 2007; 19 (10): 823-6
224. Bricout VA, Serrurier BD, Bigard AX, et al. Effects of hindlimb suspension and androgen treatment on testosterone receptors in rat skeletal muscle. Eur J Appl Physiol Occup Physiol 1999; 79 (5): 443-8
225. Deschenes MR, Maresh CM, Armstrong LE, et al. Endurance and resistance exercise induce muscle fibre type specific responses in androgen binding capacity. J Steroid Biochem 1994; 50 (3-4): 175-9
226. Matsakas A, Nikolaidis MG, Kokalas N, et al. Effect of voluntary exercise on the expression of IGF-1 and androgen receptor in three rat skeletal muscles and on serum IGF-1 and testosterone levels. Int J Sports Med 2004; 25 (7): 502-8
227. Vingren JL, Koziris LP, Gordon SE, et al. Chronic alcohol intake, resistance training, and muscle androgen receptor content. Med Sci Sports Exerc 2005; 37 (11): 1842-8
228. Willoughby DS, Taylor M, Taylor L. Glucocorticoid receptor and ubiquitin expression after repeated eccentric exercise. Med Sci Sports Exerc 2003; 35 (12): 2023-31
229. Willoughby DS. Effects of heavy resistance training on myostatin mRNA and protein expression. Med Sci Sports Exerc 2004; 36 (4): 574-82
230. Bonifazi M, Bosco C, Colli R, et al. Glucocorticoid receptors in human peripheral blood mononuclear cells in relation to explosive performance in elite handball players. Life Sci 2001; 69 (8): 961-8
231. Peijie C, Zicai D, Haowen X, et al. Effects of chronic and acute training on glucocorticoid receptors concentrations in rats. Life Sci 2004; 75 (11): 1303-11
232. Peijie B, Zicai D, Tian W. Changes in cytosol glucocorticoid receptor and renal Na(+)-K(+)-ATPase activity after swimming training in rat. J Sports Med Phys Fit 2007; 47 (2): 246-9
233. Peijie C, Renbao X, Xinming T. Long-term endurance training induced changes in glucocorticoid receptors concentrations in rat and man. J Sports Med Phys Fit 2004; 44 (3): 322-7
234. Coutinho AE, Campbell JE, Fediuc S, et al. Effect of voluntary exercise on peripheral tissue glucocorticoid receptor content and the expression and activity of 11β-HSD1 in the Syrian hamster. J Appl Physiol 2006; 100 (5): 1483-8
235. Kraemer WJ. Endocrine responses to resistance exercise. In: Baechle TR, Earle RW, editors. Essentials of strength training and conditioning. 2nd ed. Champaign (IL): Human Kinetics, 2000: 91-114
236. Hackney AC, Viru A. Research methodology: endocrinologic measurements in exercise science and sports medicine. J Athl Train 2008; 43 (6): 631-9
237. Tremblay MS, Chu SY. Hormonal responses to exercise: methodological considerations. In: Warren MP, Constantini NW, editors. Sports endocrinology. Totowa (NJ): Humana Press, Inc., 2000: 1-30
238. Kargotich S, Goodman C, Keast D, et al. Influence of exercise-induced plasma volume changes on the interpretation of biochemical data following high-intensity exercise. Clin J Sports Med 1997; 7 (3): 185-91
239. Kraemer RR, Kilgore JL, Kraemer GR, et al. Growth hormone, IGF-1, and testosterone responses to resistive exercise. Med Sci Sports Exerc 1992; 24 (12): 1346-52
240. Gupta SK, Lindemulder EA, Sathyan G. Modeling of circadian testosterone in healthy men and hypogonadal men. J Clin Pharmacol 2000; 40 (7): 731-8
241. Thuma JR, Gilders RM, Verdun M, et al. Circadian rhythm of cortisol confounds cortisol responses to exercise: implications for future research. J Appl Physiol 1995; 78 (5): 1657-64
242. Van Cauter E, Leproult R, Kupfer DJ. Effects of gender and age on the levels and circadian rhythmicity of plasma cortisol. J Clin Endocrinol Metab 1996; 81 (7): 2468-73
243. Bird SP, Tarpenning KM. Influence of circadian time structure on acute hormonal responses to a single bout of

© 2011 Adis Data Information BV. All rights reserved.

heavy-resistance exercise in weight-trained men. Chronobiol Int 2004; 21 (1): 131-46

244. Deschenes MR, Kraemer WJ, Bush JA, et al. Biorhythmic influences on functional capacity of human muscle and physiological responses. Med Sci Sports Exerc 1998; 30 (9): 1399-407

245. Kanaley JA, Weltman JY, Pieper KS, et al. Cortisol and growth hormone responses to exercise at different times of day. J Clin Endocrinol Metab 2001; 86 (6): 2881-9

246. Bishop D. An applied research model for the sport sciences. Sports Med 2008; 38 (3): 253-63

247. Kraemer WJ, Fleck SJ, Evans WJ. Strength and power training: physiological mechanisms of adaptation. Exerc Sport Sci Rev 1996; 24: 363-97

Correspondence: Dr *Blair Crewther*, Hamlyn Centre, Institute of Global Health Innovation, Bessemer Building, Imperial College, South Kensington Campus, London SW7 2AZ, UK.
E-mail: bcrewthe@imperial.ac.uk

© 2011 Adis Data Information BV. All rights reserved.